Redox Reactions

Core Theme: *"Where there is oxidation, there is always reduction — Chemistry is essentially a study of redox systems."*

1. Learning Objectives & Overview

After studying this tutorial, you will be able to:

- Identify redox reactions as a class of chemical reactions in which oxidation and reduction occur simultaneously.
- Define the terms **oxidation**, **reduction**, **oxidant (oxidising agent)**, and **reductant (reducing agent)** across classical, electronic, and oxidation number concepts.
- Explain the mechanism of redox reactions through electron transfer processes and competitive electron transfer.
- Determine the oxidation number of elements in compounds and ions, including cases with apparent fractional oxidation states, and apply **Stock notation**.
- Classify redox reactions into **combination**, **decomposition**, **displacement**, and **disproportionation** reactions.
- Balance redox chemical equations systematically using both the **Oxidation Number Method** and the **Half-Reaction (Ion-Electron) Method** in acidic and basic media.
- Understand redox titrations, self-indicators, and iodometric/iodimetric indicator principles.
- Relate redox reactions to electrode processes, Daniell cell operation, standard electrode potentials (E°), and the electrochemical series.

Transformation of matter from one kind into another occurs through various types of reactions. One crucial category is **Redox Reactions**, which govern countless physical and biological phenomena. Their applications span pharmaceutical, biological, industrial, metallurgical, and agricultural fields. Examples include burning fuels for domestic, transport, and commercial energy; electrochemical extraction of reactive metals and non-metals; manufacturing compounds like caustic soda (NaOH); operation of dry and wet batteries; and corrosion of metals. Contemporary environmental topics like the **Hydrogen Economy** (using liquid hydrogen as fuel) and the development of the **Ozone Hole** also fall under redox phenomena.

2. Classical Idea of Redox Reactions

2.1 Classical Definition of Oxidation

Originally, the term **oxidation** described the addition of oxygen to an element or a compound. Because dioxygen is abundant in the atmosphere (~20%), many elements combine with it and commonly occur on Earth as their oxides:

$$2\mathrm{Mg(s)} + \mathrm{O_2(g)} \rightarrow 2\mathrm{MgO(s)}$$
$$\mathrm{S(s)} + \mathrm{O_2(g)} \rightarrow \mathrm{SO_2(g)}$$

Here, magnesium and sulphur are oxidised by the addition of oxygen. Similarly, methane is oxidised due to the addition of oxygen:

$$\mathrm{CH_4(g)} + 2\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + 2\mathrm{H_2O(l)}$$

In the combustion of methane, hydrogen is replaced by oxygen. This prompted chemists to broaden the scope of oxidation to include the **removal of hydrogen** from a substance, as seen in the oxidation of hydrogen sulphide:

$$2\mathrm{H_2S(g)} + \mathrm{O_2(g)} \rightarrow 2\mathrm{S(s)} + 2\mathrm{H_2O(l)}$$

Chemists further extended oxidation to reactions involving other electronegative elements besides oxygen (such as fluorine, chlorine, and sulphur):

$$\mathrm{Mg(s)} + \mathrm{F_2(g)} \rightarrow \mathrm{MgF_2(s)}$$
$$\mathrm{Mg(s)} + \mathrm{Cl_2(g)} \rightarrow \mathrm{MgCl_2(s)}$$

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$$\mathrm{Mg(s) + S(s) \rightarrow MgS(s)}$$

Finally, the removal of electropositive elements was also recognised as oxidation. For instance, the conversion of potassium ferrocyanide to potassium ferricyanide involves the removal of the electropositive element potassium:

$$\mathrm{2K_4[Fe(CN)_6](aq) + H_2O_2(aq) \rightarrow 2K_3[Fe(CN)_6](aq) + 2KOH(aq)}$$

2.2 Classical Definition of Reduction

Initially, reduction meant the removal of oxygen from a compound. Today, it is broadened to include the removal of oxygen or any electronegative element, or the addition of hydrogen or any electropositive element:

- **Removal of oxygen from mercuric oxide:**
$$\mathrm{2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)}$$
- **Removal of electronegative element (chlorine) from ferric chloride:**
$$\mathrm{2FeCl_3(aq) + H_2(g) \rightarrow 2FeCl_2(aq) + 2HCl(aq)}$$
- **Addition of hydrogen to ethene:**
$$\mathrm{CH_2 = CH_2(g) + H_2(g) \rightarrow H_3C - CH_3(g)}$$
- **Addition of electropositive element (mercury) to mercuric chloride:**
$$\mathrm{2HgCl_2(aq) + SnCl_2(aq) \rightarrow Hg_2Cl_2(s) + SnCl_4(aq)}$$
Note: Simultaneously, stannous chloride ($\mathrm{SnCl_2}$) is oxidised to stannic chloride ($\mathrm{SnCl_4}$) because the electronegative element chlorine is added to it. Because oxidation and reduction always occur together, the term "**redox**" was coined.

Summary of Classical Definitions

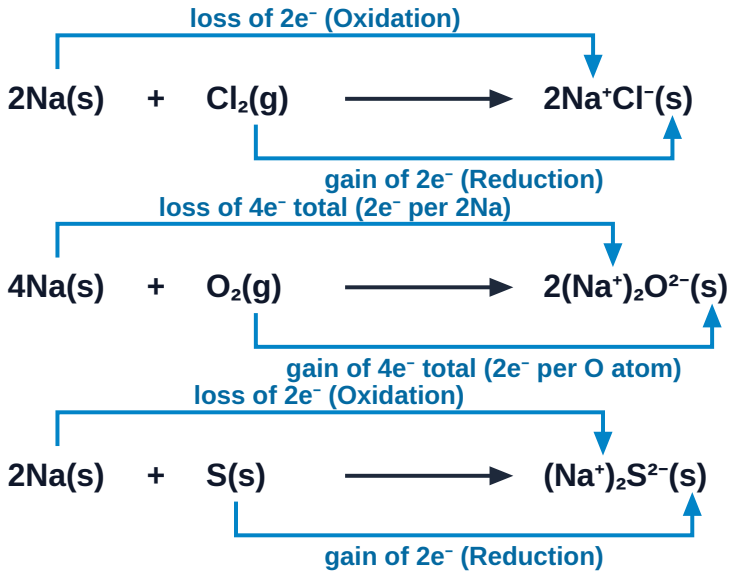
- **Oxidation:** Addition of oxygen / electronegative element to a substance OR removal of hydrogen / electropositive element from a substance.
- **Reduction:** Removal of oxygen / electronegative element from a substance OR addition of hydrogen / electropositive element to a substance.

3. Redox Reactions in Terms of Electron Transfer

Consider the formation of ionic compounds such as sodium chloride $\mathrm{Na^+Cl^-(s)}$, sodium oxide $\mathrm{(Na^+)_2O^{2-}(s)}$, and sodium sulphide $\mathrm{(Na^+)_2S^{2-}(s)}$:

$$\mathrm{2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)}$$
$$\mathrm{4Na(s) + O_2(g) \rightarrow 2Na_2O(s)}$$
$$\mathrm{2Na(s) + S(s) \rightarrow Na_2S(s)}$$

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Tutorial Diagram 1: Electron transfer representation in the formation of sodium chloride, sodium oxide, and sodium sulphide.

Each process can be split into two separate steps called **half-reactions** that explicitly show the involvement of electrons. For example, in the formation of sodium chloride:

- **Oxidation half-reaction (loss of electrons):** $2\mathrm{Na(s)} \rightarrow 2\mathrm{Na^+(g)} + 2\mathrm{e^-}$
- **Reduction half-reaction (gain of electrons):** $\mathrm{Cl_2(g)} + 2\mathrm{e^-} \rightarrow 2\mathrm{Cl^-(g)}$
- **Overall redox reaction:** $2\mathrm{Na(s)} + \mathrm{Cl_2(g)} \rightarrow 2\mathrm{Na^+Cl^-(s)}$ or $2\mathrm{NaCl(s)}$

Electronic Definitions

- **Oxidation:** Loss of electron(s) by any species.
- **Reduction:** Gain of electron(s) by any species.
- **Oxidising agent (Oxidant):** Acceptor of electron(s) (undergoes reduction itself).
- **Reducing agent (Reductant):** Donor of electron(s) (undergoes oxidation itself).

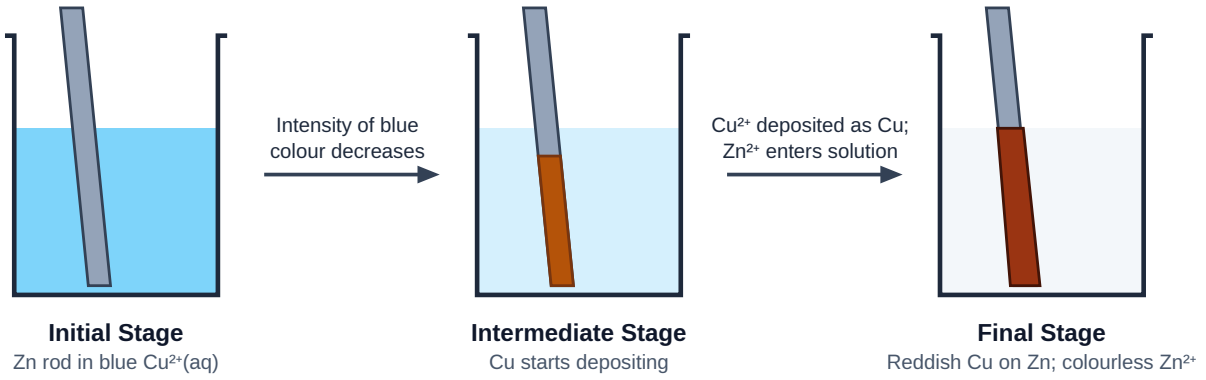
3.1 Competitive Electron Transfer Reactions

1. Zinc in Copper Nitrate Solution: When a strip of metallic zinc is placed in an aqueous blue solution of copper nitrate $\mathrm{Cu(NO_3)_2}$ for about one hour, the zinc strip becomes coated with reddish metallic copper and the blue colour of $\mathrm{Cu^{2+}}$ disappears. Colourless $\mathrm{Zn^{2+}}$ ions form in solution (confirmed by passing $\mathrm{H_2S}$ gas into the ammoniacal solution to get a white precipitate of ZnS).

$$\mathrm{Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)}$$

Here, Zn loses $2\mathrm{e^-}$ to form $\mathrm{Zn^{2+}}$ (oxidation) and $\mathrm{Cu^{2+}}$ gains $2\mathrm{e^-}$ to form Cu (reduction). If we place metallic copper in a zinc sulphate ($\mathrm{ZnSO_4}$) solution, no visible reaction occurs, and passing $\mathrm{H_2S}$ gas produces no black precipitate of cupric sulphide (CuS), despite CuS having extremely low solubility. Thus, the equilibrium greatly favours the products $\mathrm{Zn^{2+}(aq)}$ and $\mathrm{Cu(s)}$.

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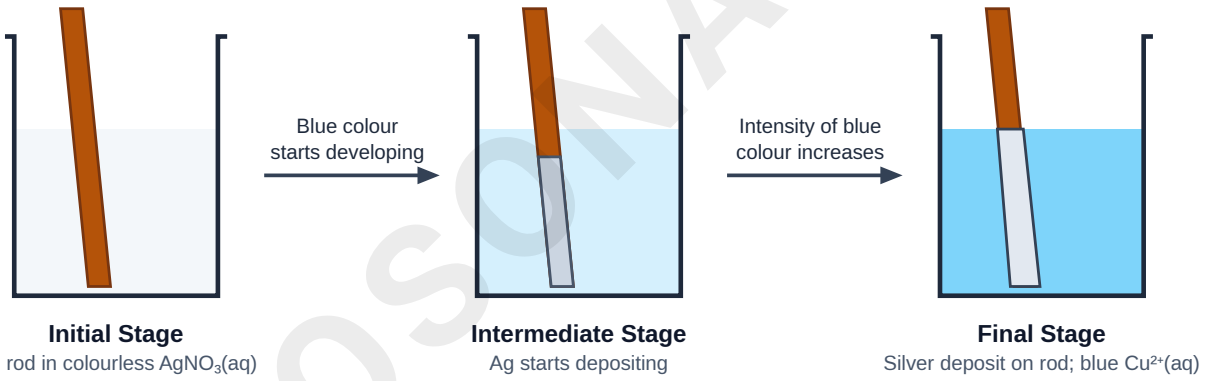


Tutorial Diagram 2: Redox reaction between metallic zinc and an aqueous solution of copper nitrate.

2. Copper in Silver Nitrate Solution: When a copper rod is dipped into a colourless aqueous silver nitrate ($\mathrm{AgNO_3}$) solution, silver deposits on the copper rod and the solution develops a blue colour due to the formation of $\mathrm{Cu^{2+}(aq)}$ ions:

$$\mathrm{Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)}$$

Here, $\mathrm{Cu(s)}$ loses $\mathrm{2e^-}$ to form $\mathrm{Cu^{2+}(aq)}$ and $\mathrm{2Ag^{+}(aq)}$ gains $\mathrm{2e^-}$ to form $\mathrm{2Ag(s)}$. Equilibrium greatly favours the products $\mathrm{Cu^{2+}(aq)}$ and $\mathrm{Ag(s)}$.



Tutorial Diagram 3: Redox reaction between metallic copper and an aqueous solution of silver nitrate.

3. Cobalt in Nickel Sulphate Solution: When metallic cobalt is placed in nickel sulphate ($\mathrm{NiSO_4}$) solution:

$$\mathrm{Co(s) + Ni^{2+}(aq) \rightarrow Co^{2+}(aq) + Ni(s)}$$

At equilibrium, chemical tests reveal that both $\mathrm{Ni^{2+}(aq)}$ and $\mathrm{Co^{2+}(aq)}$ are present at moderate concentrations. Neither reactants nor products are overwhelmingly favoured.

Conclusion on Metal Activity: From these comparisons, the electron-releasing tendency (reducing activity) of the metals follows the order: $\mathrm{Zn > Cu > Ag}$. Listing metals and their ions based on their tendency to release electrons gives the **metal activity series** or **electrochemical series**, which helps design **Galvanic cells**.

4. Concept of Oxidation Number

In covalent reactions such as $\mathrm{2H_2(g) + O_2(g) \rightarrow 2H_2O(l)}$, $\mathrm{H_2(g) + Cl_2(g) \rightarrow 2HCl(g)}$, and $\mathrm{CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)}$, charge transfer is only partial (an electron shift rather than complete loss or gain). To keep track of electron shifts, chemists use the **oxidation number** method.

> **Definition of Oxidation Number:** Oxidation number denotes the oxidation state of an element in a compound ascertained according to a set of rules formulated on the basis that the electron pair in a covalent bond belongs

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entirely to the more electronegative element.

4.1 Rules for Assigning Oxidation Numbers

1. **Elements in Free/Uncombined State:** Each atom in an uncombined element has an oxidation number of **zero** (0). Examples: $\mathrm{H_2}, \mathrm{O_2}, \mathrm{Cl_2}, \mathrm{O_3}, \mathrm{P_4}, \mathrm{S_8}, \mathrm{Na}, \mathrm{Mg}, \mathrm{Al}$.
2. **Monoatomic Ions & Fixed-Valency Metals:** For ions composed of only one atom, the oxidation number equals the charge on the ion (e.g., $\mathrm{Na^+} = +1, \mathrm{Mg^{2+}} = +2, \mathrm{Fe^{3+}} = +3, \mathrm{Cl^-} = -1, \mathrm{O^{2-}} = -2$). In all their compounds:
   - All **alkali metals (Group 1)** have an oxidation number of $+1$.
   - All **alkaline earth metals (Group 2)** have an oxidation number of $+2$.
   - **Aluminium** (Al) is regarded to have an oxidation number of $+3$ in all its compounds.
3. **Oxygen:** In most compounds, oxygen has an oxidation number of -2. There are two important classes of exceptions:
   - **Peroxides and Superoxides (O–O direct linkage):** In peroxides (e.g., $\mathrm{H_2O_2}, \mathrm{Na_2O_2}, \mathrm{CaO_2}$), each oxygen atom has an oxidation number of -1. In superoxides (e.g., $\mathrm{KO_2}, \mathrm{RbO_2}$), each oxygen atom has an oxidation number of $-1/2$.
   - **Oxygen bonded to Fluorine:** Because fluorine is more electronegative than oxygen, oxygen exhibits positive oxidation states: $+2$ in oxygen difluoride $(\mathrm{OF_2})$ and $+1$ in dioxygen difluoride $(\mathrm{O_2F_2})$.
4. **Hydrogen:** The oxidation number of hydrogen is $+1$, except when it is bonded to metals in binary compounds (ionic metal hydrides such as $\mathrm{LiH}, \mathrm{NaH}, \mathrm{CaH_2}$), where its oxidation number is -1.
5. **Halogens:**
   - **Fluorine** (F) is the most electronegative element and has an oxidation number of -1 in *all* its compounds.
   - Other halogens $(\mathrm{Cl}, \mathrm{Br}, \mathrm{I})$ have an oxidation number of -1 when they occur as halide ions in their compounds, but exhibit **positive oxidation numbers** when combined with oxygen in oxoacids and oxoanions.
6. **Algebraic Sum Rule:**
   - In a neutral compound, the algebraic sum of the oxidation numbers of all atoms equals **zero** (0).
   - In a polyatomic ion, the algebraic sum of the oxidation numbers of all atoms equals the **net charge on the ion** (e.g., in $\mathrm{CO_3^{2-}}$, the sum is -2).

4.2 Periodic Trends & Stock Notation

Metallic elements have positive oxidation numbers, whereas non-metallic elements can have positive or negative oxidation numbers. For representative elements, the **highest oxidation number** is the group number for the first two groups (Groups 1 and 2) and **(Group number** $-$ **10)** for Groups 13 to 17.

Group	1	2	13	14	15	16	17
Element (Period 3)	Na	Mg	Al	Si	P	S	Cl
Representative Compound	NaCl	$\mathrm{MgSO_4}$	$\mathrm{AlF_3}$	$\mathrm{SiCl_4}$	$\mathrm{P_4O_{10}}$	$\mathrm{SF_6}$	$\mathrm{HClO_4}$
Highest Oxidation State	$+1$	$+2$	$+3$	$+4$	$+5$	$+6$	$+7$

Stock Notation: Proposed by German chemist Alfred Stock, the oxidation state of a metal in a compound is represented by a Roman numeral in parentheses immediately after the symbol of the metal in the molecular formula. Examples: aurous chloride is $\mathrm{Au(I)Cl}$, auric chloride is $\mathrm{Au(III)Cl_3}$, stannous chloride is $\mathrm{Sn(II)Cl_2}$, stannic chloride is $\mathrm{Sn(IV)Cl_4}$, and mercurous chloride $\mathrm{Hg_2(I)Cl_2}$ is the reduced form of mercuric chloride $\mathrm{Hg(II)Cl_2}$.

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Redox Definitions in Terms of Oxidation Number

- **Oxidation:** An increase in the oxidation number of the element in the given substance.
- **Reduction:** A decrease in the oxidation number of the element in the given substance.
- **Oxidising agent (Oxidant):** A reagent which can increase the oxidation number of an element in a given substance (its own oxidation number decreases).
- **Reducing agent (Reductant):** A reagent which lowers the oxidation number of an element in a given substance (its own oxidation number increases).
- **Redox reactions:** Reactions which involve a change in the oxidation number of the interacting species.

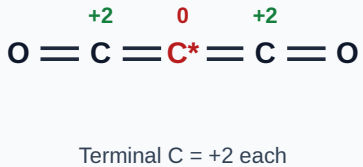
4.3 The Paradox of Fractional Oxidation Number & Peroxy Acids

Sometimes, the calculated oxidation number of an element in a compound is a fraction, such as:

- C_3O_2 (carbon suboxide): Average oxidation number of $C = +4/3$.
- Br_3O_8 (tribromooctaoxide): Average oxidation number of $Br = +16/3$.
- $Na_2S_4O_6$ or $S_4O_6^{2-}$ (tetrathionate ion): Average oxidation number of $S = +2.5$ (or $+5/2$).

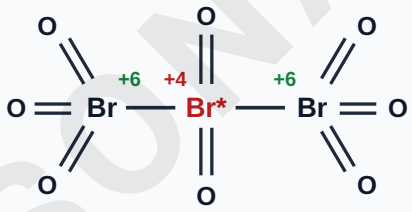
Because electrons are never shared or transferred in fractions, a fractional oxidation state is merely the **average oxidation state** of the atoms of that element in the molecule. Chemical structures reveal that the atoms are present in different whole-number oxidation states:

Carbon Suboxide (C_3O_2)



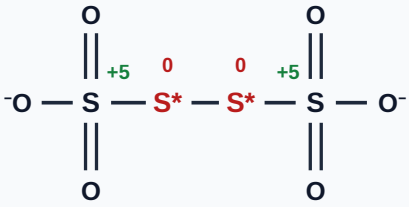
Terminal C = +2 each
Middle C* = 0
Average = +4/3

Tribromooctaoxide (Br_3O_8)



Terminal Br = +6, Middle Br* = +4
Average = +16/3

Tetrathionate Ion ($S_4O_6^{2-}$)



Terminal S = +5, Middle S* = 0
Average = +2.5

Tutorial Diagram 4: Structural resolution of apparent fractional oxidation states in C_3O_2, Br_3O_8, and $S_4O_6^{2-}$.

Other examples of apparent fractional oxidation numbers include mixed oxides: Fe_3O_4 ($FeO \cdot Fe_2O_3$, average $+8/3$), Mn_3O_4 ($MnO \cdot Mn_2O_3$, average $+8/3$), and Pb_3O_4 ($2PbO \cdot PbO_2$, average $+8/3$). Similarly, in KI_3, the I_3^- ion consists of an I_2 molecule (0) coordinated to an I^- ion (-1), giving an average of $-1/3$. True fractional oxidation numbers occur in diatomic ions where electrons are delocalised equally over two identical atoms, such as O_2^+ ($+1/2$) and O_2^- ($-1/2$).

Fallacy of Blind Algebraic Calculation (Peroxy Linkages): In compounds like Caro's acid (H_2SO_5) and chromium pentoxide (CrO_5), blind application of the algebraic formula (assuming all oxygens are -2) gives $+8$ for S in H_2SO_5 and $+10$ for Cr in CrO_5. Both are impossible because the maximum oxidation state cannot exceed the number of valence electrons ($+6$ for both S and Cr):

- In H_2SO_5 ($H-O-O-SO_2-OH$), two oxygen atoms are in a peroxide linkage (-1 each) and three are normal oxide atoms (-2 each): $2(+1) + x + 2(-1) + 3(-2) = 0 \implies x = +6$.
- In $Cr_2O_7^{2-}$ (dichromate ion), there is no peroxide linkage; each Cr is in its highest $+6$ state. (In butterfly-structured CrO_5, four oxygens are in two peroxo rings at -1 each and one double-bonded oxygen is at -2, giving $Cr = +6$).

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- In $\mathrm{NO_3^-}$ (nitrate ion), nitrogen has one coordinate bond, one double bond, and one single bond to oxygen, giving a true oxidation number of $+5$.

5. Classification of Redox Reactions

5.1 Combination Reactions ($\mathrm{A + B \rightarrow C}$)

For a combination reaction to be a redox reaction, either A or B or both A and B must be in the elemental form. All combustion reactions utilizing elemental $\mathrm{O_2}$ belong here:

$$\overset{0}{\mathrm{C}}(\mathrm{s}) + \overset{0}{\mathrm{O_2}}(\mathrm{g}) \xrightarrow{\Delta} \overset{+4\,-2}{\mathrm{CO_2}}(\mathrm{g})$$

$$3\overset{0}{\mathrm{Mg}}(\mathrm{s}) + \overset{0}{\mathrm{N_2}}(\mathrm{g}) \xrightarrow{\Delta} \overset{+2\ \ -3}{\mathrm{Mg_3N_2}}(\mathrm{s})$$

$$\overset{-4\,+1}{\mathrm{CH_4}}(\mathrm{g}) + 2\overset{0}{\mathrm{O_2}}(\mathrm{g}) \xrightarrow{\Delta} \overset{+4\,-2}{\mathrm{CO_2}}(\mathrm{g}) + 2\overset{+1\,-2}{\mathrm{H_2O}}(\mathrm{l})$$

5.2 Decomposition Reactions

Decomposition is the opposite of combination: a compound breaks down into two or more components, **at least one of which must be in the elemental state**:

$$2\overset{+1\,-2}{\mathrm{H_2O}}(\mathrm{l}) \xrightarrow{\Delta} 2\overset{0}{\mathrm{H_2}}(\mathrm{g}) + \overset{0}{\mathrm{O_2}}(\mathrm{g})$$

$$2\overset{+1\,-1}{\mathrm{NaH}}(\mathrm{s}) \xrightarrow{\Delta} 2\overset{0}{\mathrm{Na}}(\mathrm{s}) + \overset{0}{\mathrm{H_2}}(\mathrm{g})$$

$$2\overset{+1+5\,-2}{\mathrm{KClO_3}}(\mathrm{s}) \xrightarrow{\Delta} 2\overset{+1-1}{\mathrm{KCl}}(\mathrm{s}) + 3\overset{0}{\mathrm{O_2}}(\mathrm{g})$$

Crucial Exam Alert: Not all decomposition reactions are redox reactions! For example, the thermal decomposition of calcium carbonate involves no change in oxidation numbers and is *not* a redox reaction:

$$\overset{+2\,+4\,-2}{\mathrm{CaCO_3}}(\mathrm{s}) \xrightarrow{\Delta} \overset{+2\,-2}{\mathrm{CaO}}(\mathrm{s}) + \overset{+4\,-2}{\mathrm{CO_2}}(\mathrm{g})$$

5.3 Displacement Reactions ($\mathrm{X + YZ \rightarrow XZ + Y}$)

An ion (or atom) in a compound is replaced by an ion (or atom) of another element.

(a) Metal Displacement

A metal in a compound is displaced by a stronger reducing metal in the uncombined state. Used widely in metallurgy:

$$\overset{+2}{\mathrm{CuSO_4}}(\mathrm{aq}) + \overset{0}{\mathrm{Zn}}(\mathrm{s}) \rightarrow \overset{0}{\mathrm{Cu}}(\mathrm{s}) + \overset{+2}{\mathrm{ZnSO_4}}(\mathrm{aq})$$

$$\overset{+5}{\mathrm{V_2O_5}}(\mathrm{s}) + 5\overset{0}{\mathrm{Ca}}(\mathrm{s}) \xrightarrow{\Delta} 2\overset{0}{\mathrm{V}}(\mathrm{s}) + 5\overset{+2}{\mathrm{CaO}}(\mathrm{s})$$

$$\overset{+4}{\mathrm{TiCl_4}}(\mathrm{l}) + 2\overset{0}{\mathrm{Mg}}(\mathrm{s}) \xrightarrow{\Delta} \overset{0}{\mathrm{Ti}}(\mathrm{s}) + 2\overset{+2}{\mathrm{MgCl_2}}(\mathrm{s})$$

$$\overset{+3}{\mathrm{Cr_2O_3}}(\mathrm{s}) + 2\overset{0}{\mathrm{Al}}(\mathrm{s}) \xrightarrow{\Delta} \overset{+3}{\mathrm{Al_2O_3}}(\mathrm{s}) + 2\overset{0}{\mathrm{Cr}}(\mathrm{s})$$

(b) Non-Metal Displacement

- **Displacement of Hydrogen from Cold Water:** All alkali metals and active alkaline earth metals ($\mathrm{Ca, Sr, Ba}$) displace $\mathrm{H_2}$ from cold water:

$$2\mathrm{Na}(\mathrm{s}) + 2\mathrm{H_2O}(\mathrm{l}) \rightarrow 2\mathrm{NaOH}(\mathrm{aq}) + \mathrm{H_2}(\mathrm{g})$$

$$\mathrm{Ca}(\mathrm{s}) + 2\mathrm{H_2O}(\mathrm{l}) \rightarrow \mathrm{Ca(OH)_2}(\mathrm{aq}) + \mathrm{H_2}(\mathrm{g})$$

- **Displacement of Hydrogen from Steam:** Less active metals like Mg and Fe react with steam:

$$\mathrm{Mg}(\mathrm{s}) + 2\mathrm{H_2O}(\mathrm{l}) \xrightarrow{\Delta} \mathrm{Mg(OH)_2}(\mathrm{s}) + \mathrm{H_2}(\mathrm{g})$$

$$2\mathrm{Fe}(\mathrm{s}) + 3\mathrm{H_2O}(\mathrm{l}) \xrightarrow{\Delta} \mathrm{Fe_2O_3}(\mathrm{s}) + 3\mathrm{H_2}(\mathrm{g})$$

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- **Displacement of Hydrogen from Acids:** Metals like $\mathrm{Zn}, \mathrm{Mg}, \mathrm{Fe}$ (and even Cd and Sn, which do not react with steam) displace $\mathrm{H_2}$ from dilute acids ($\mathrm{Mg}, \mathrm{Zn}, \mathrm{Fe}$ with HCl give $\mathrm{MgCl_2}, \mathrm{ZnCl_2}, \mathrm{FeCl_2}$ and $\mathrm{H_2}$). Rate of $\mathrm{H_2}$ evolution reflects reactivity: fastest for Mg, slowest for Fe. Noble metals like Ag and Au do not react with HCl.
- **Halogen Displacement & Oxidising Power:** Oxidising power of halogens decreases down Group 17: $\mathrm{F_2} > \mathrm{Cl_2} > \mathrm{Br_2} > \mathrm{I_2}$.
  - $\mathrm{F_2}$ is so reactive that it attacks water to displace oxygen: $\mathrm{2H_2O(l) + 2F_2(g) \rightarrow 4HF(aq) + O_2(g)}$. Because $\mathrm{F_2}$ is the strongest oxidising agent, $\mathrm{F^-}$ ions cannot be oxidised to $\mathrm{F_2}$ by any chemical reagent—only electrolytically!
  - $\mathrm{Cl_2}$ displaces $\mathrm{Br^-}$ and $\mathrm{I^-}$ in aqueous solution; $\mathrm{Br_2}$ displaces $\mathrm{I^-}$:

$$\mathrm{Cl_2(g) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(l)}$$
$$\mathrm{Cl_2(g) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(s)}$$
$$\mathrm{Br_2(l) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(s)}$$

  Since $\mathrm{Br_2}$ and $\mathrm{I_2}$ are coloured and dissolve in $\mathrm{CCl_4}$, these reactions form the basis of the **'Layer Test'** in qualitative analysis.

5.4 Disproportionation Reactions

In a **disproportionation reaction**, the *same element in one oxidation state is simultaneously oxidised and reduced*. The reacting substance must contain an element that can exist in at least three oxidation states, and the reacting species must be in an **intermediate oxidation state**.

- **Decomposition of Hydrogen Peroxide (O in -1 state):**

$$\mathrm{\overset{+1\ -1}{2H_2O_2(aq) \rightarrow 2\overset{+1\ -2}{H_2O}(l) + \overset{0}{O_2}(g)}}$$

- **White Phosphorus in Alkaline Medium (P in 0 state):**

$$\mathrm{\overset{0}{P_4}(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow \overset{-3}{PH_3}(g) + 3H_2\overset{+1}{P}O_2^-(aq)}$$

- **Sulphur in Alkaline Medium (S in 0 state):**

$$\mathrm{\overset{0}{S_8}(s) + 12OH^-(aq) \rightarrow 4\overset{-2}{S^{2-}}(aq) + 2\overset{+2}{S_2}O_3^{2-}(aq) + 6H_2O(l)}$$

- **Chlorine in Alkaline Medium (Cl in 0 state — formation of household bleach $\mathrm{ClO^-}$):**

$$\mathrm{\overset{0}{Cl_2}(g) + 2OH^-(aq) \rightarrow \overset{+1}{ClO^-}(aq) + \overset{-1}{Cl^-}(aq) + H_2O(l)}$$

- **Why Fluorine NEVER Disproportionates:** Being the most electronegative element, fluorine cannot exhibit any positive oxidation state. With alkali, it forms $\mathrm{F^-}$ (-1) and $\mathrm{OF_2}$ (-1):

$$\mathrm{2F_2(g) + 2OH^-(aq) \rightarrow 2F^-(aq) + OF_2(g) + H_2O(l)}$$

6. Balancing Redox Reactions & Redox Titrations

6.1 Method 1: Oxidation Number Method

1. Write the correct formula for each reactant and product.
2. Assign oxidation numbers to all elements and identify the atoms undergoing oxidation and reduction.
3. Calculate the increase or decrease in oxidation number per atom and for the entire molecule/ion. Multiply by suitable coefficients to make total increase equal to total decrease.
4. Balance ionic charges by adding $\mathrm{H^+}$ ions (in acidic medium) or $\mathrm{OH^-}$ ions (in basic medium) on the appropriate side.
5. Balance hydrogen atoms by adding $\mathrm{H_2O}$ molecules, then verify that oxygen atoms are automatically balanced.

6.2 Method 2: Half-Reaction (Ion-Electron) Method

1. Write the unbalanced equation in ionic form and split it into an **oxidation half-reaction** and a **reduction half-reaction**.
2. In each half-reaction, balance atoms other than O and H.
3. To balance O and H in **acidic medium**: add $\mathrm{H_2O}$ to the side deficient in O, then add $\mathrm{H^+}$ to the side deficient in H.
4. If in **basic medium**: first balance as in acidic medium, then add an equal number of $\mathrm{OH^-}$ ions to *both sides* for every $\mathrm{H^+}$ ion present. Combine $\mathrm{H^+}$ and $\mathrm{OH^-}$ on the same side to form $\mathrm{H_2O}$ and cancel excess water

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molecules.

5. Add electrons (e^-) to balance the net charge in each half-reaction. Multiply the half-reactions by suitable integers to equalise the number of electrons lost and gained, then add them together.

Worked Illustration: Balancing Fe^{2+} and $Cr_2O_7^{2-}$ in Acidic Medium

Skeletal equation: $Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) \rightarrow Fe^{3+}(aq) + Cr^{3+}(aq)$

- **Oxidation half:** $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$ *(Multiply by 6)* $\implies 6Fe^{2+}(aq) \rightarrow 6Fe^{3+}(aq) + 6e^-$
- **Reduction half:** $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$
- **Balanced net ionic equation:**

$$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

6.3 Redox Reactions as the Basis for Titrations

In redox titrations, the strength of a reductant or oxidant is determined using a redox-sensitive indicator:

1. **Self-Indicator (MnO_4^-):** Potassium permanganate (MnO_4^-) is intensely coloured. When titrating reductants like Fe^{2+} or oxalate ($C_2O_4^{2-}$), MnO_4^- acts as its own indicator. A lasting tinge of pink appears immediately after the equivalence point at MnO_4^- concentrations as low as $10^{-6}\,mol\,dm^{-3}$ ($10^{-6}\,mol\,L^{-1}$).
2. **Redox-Sensitive External Indicator ($Cr_2O_7^{2-}$):** Dichromate ($Cr_2O_7^{2-}$) is not a self-indicator. Instead, it oxidises the indicator **diphenylamine** immediately after all the reductant is consumed, producing an intense blue colour at the end point.
3. **Iodometric Titration (Starch Indicator):** Oxidising agents capable of oxidising I^- ions (such as Cu^{2+}) liberate molecular iodine (I_2, which remains dissolved in KI solution as KI_3):

$$2Cu^{2+}(aq) + 4I^-(aq) \rightarrow Cu_2I_2(s) + I_2(aq)$$

The liberated I_2 gives an **intense blue colour with starch** and is titrated against standard sodium thiosulphate ($S_2O_3^{2-}$) solution:

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

As soon as all I_2 is consumed by $S_2O_3^{2-}$, the blue colour disappears, marking the sharp end point.

6.4 Limitations of the Concept of Oxidation Number

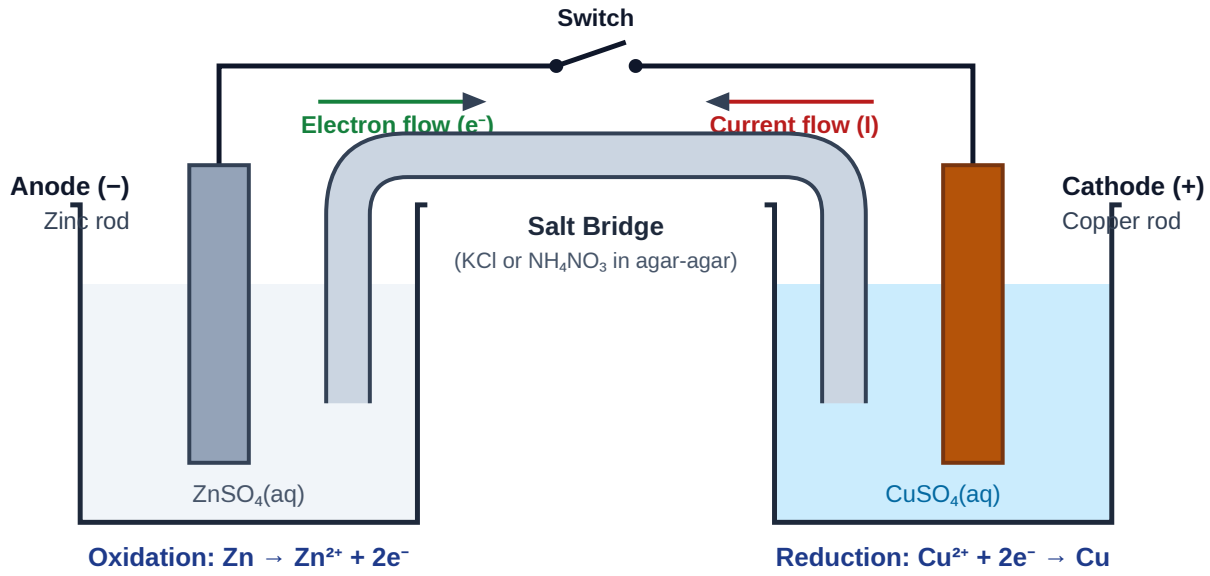
In modern chemistry, the concept of redox processes has evolved further: **oxidation** is visualised as a *decrease in electron density* around the atom(s) involved, while **reduction** is visualised as an *increase in electron density* around the atom(s).

7. Redox Reactions and Electrode Processes

When a zinc rod is dipped directly into a $CuSO_4$ solution, electrons transfer directly from Zn to Cu^{2+} and heat is evolved. In a **Daniell Cell**, we separate the two half-reactions into two beakers so that electron transfer occurs *indirectly* through an external metallic wire:

- **Left Beaker (Anode Compartment):** A zinc rod dipped in $ZnSO_4(aq)$ solution. Oxidation occurs: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$.
- **Right Beaker (Cathode Compartment):** A copper rod dipped in $CuSO_4(aq)$ solution. Reduction occurs: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$.
- **Redox Couple:** Defined as having together the oxidised and reduced forms of a substance taking part in an oxidation or reduction half-reaction, separated by a slash ($/$) representing an interface (oxidised form written first): Zn^{2+}/Zn and Cu^{2+}/Cu.
- **Salt Bridge:** A U-tube filled with a solution of KCl or NH_4NO_3 solidified with agar-agar into a jelly. It provides electrical contact and maintains electrical neutrality by ion migration without allowing the two solutions to mix.

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Tutorial Diagram 5: Set-up of a Daniell cell showing indirect electron transfer from the zinc anode to the copper cathode and ion migration through the salt bridge.

7.1 Standard Electrode Potentials (E°) at $298\ \mathrm{K}$

When the concentration of each species taking part in an electrode reaction is unity ($1\ \mathrm{M}$) and any gas involved is at $1\ \mathrm{atm}$ pressure at $298\ \mathrm{K}$, the potential of the electrode is called the **Standard Electrode Potential** (E°). By convention, E° of the standard hydrogen electrode ($\mathrm{H^+/H_2}$) is $0.00\ \mathrm{V}$.

- A **negative** E° means the redox couple is a **stronger reducing agent** than the $\mathrm{H^+/H_2}$ couple.
- A **positive** E° means the redox couple is a **weaker reducing agent** than the $\mathrm{H^+/H_2}$ couple (and its oxidised form is a stronger oxidising agent than $\mathrm{H^+}$).

Reduction Half-Reaction (Oxidised form $+ \mathrm{ne^-} \rightarrow$ Reduced form**)**	E°/V	**Reduction Half-Reaction (** Oxidised form $+ \mathrm{ne^-} \rightarrow$ Reduced form**)**	E°/V
$\mathrm{F_2(g) + 2e^- \rightarrow 2F^-}$	$+2.87$	$\mathrm{Cu^{2+} + 2e^- \rightarrow Cu(s)}$	$+0.34$
$\mathrm{Co^{3+} + e^- \rightarrow Co^{2+}}$	$+1.81$	$\mathrm{AgCl(s) + e^- \rightarrow Ag(s) + Cl^-}$	$+0.22$
$\mathrm{H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O}$	$+1.78$	$\mathrm{AgBr(s) + e^- \rightarrow Ag(s) + Br^-}$	$+0.10$
$\mathrm{MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O}$	$+1.51$	$\mathrm{2H^+ + 2e^- \rightarrow H_2(g)}$	0.00
$\mathrm{Au^{3+} + 3e^- \rightarrow Au(s)}$	$+1.40$	$\mathrm{Pb^{2+} + 2e^- \rightarrow Pb(s)}$	-0.13
$\mathrm{Cl_2(g) + 2e^- \rightarrow 2Cl^-}$	$+1.36$	$\mathrm{Sn^{2+} + 2e^- \rightarrow Sn(s)}$	-0.14
$\mathrm{Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O}$	$+1.33$	$\mathrm{Ni^{2+} + 2e^- \rightarrow Ni(s)}$	-0.25
$\mathrm{O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O}$	$+1.23$	$\mathrm{Fe^{2+} + 2e^- \rightarrow Fe(s)}$	-0.44
$\mathrm{MnO_2(s) + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O}$	$+1.23$	$\mathrm{Cr^{3+} + 3e^- \rightarrow Cr(s)}$	-0.74
$\mathrm{Br_2 + 2e^- \rightarrow 2Br^-}$	$+1.09$	$\mathrm{Zn^{2+} + 2e^- \rightarrow Zn(s)}$	-0.76
$\mathrm{NO_3^- + 4H^+ + 3e^- \rightarrow NO(g) + 2H_2O}$	$+0.97$	$\mathrm{2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-}$	-0.83
$\mathrm{2Hg^{2+} + 2e^- \rightarrow Hg_2^{2+}}$	$+0.92$	$\mathrm{Al^{3+} + 3e^- \rightarrow Al(s)}$	-1.66
$\mathrm{Ag^+ + e^- \rightarrow Ag(s)}$	$+0.80$	$\mathrm{Mg^{2+} + 2e^- \rightarrow Mg(s)}$	-2.36
$\mathrm{Fe^{3+} + e^- \rightarrow Fe^{2+}}$	$+0.77$	$\mathrm{Na^+ + e^- \rightarrow Na(s)}$	-2.71
$\mathrm{O_2(g) + 2H^+ + 2e^- \rightarrow H_2O_2}$	$+0.68$	$\mathrm{Ca^{2+} + 2e^- \rightarrow Ca(s)}$	-2.87
$\mathrm{I_2(s) + 2e^- \rightarrow 2I^-}$	$+0.54$	$\mathrm{K^+ + e^- \rightarrow K(s)}$	-2.93
$\mathrm{Cu^+ + e^- \rightarrow Cu(s)}$	$+0.52$	$\mathrm{Li^+ + e^- \rightarrow Li(s)}$	-3.05

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8. Comprehensive Worked Examples & Conceptual Applications

Worked Example 1: Assigning Oxidation Numbers & Stock Formulas

Problem: (I) Determine the oxidation number of the bold element in: (a) $\mathrm{NaH_2\mathbf{P}O_4}$, (b) $\mathrm{H_4\mathbf{P_2}O_7}$, (c) $\mathrm{K_2\mathbf{Mn}O_4}$, (d) $\mathrm{Ca\mathbf{O_2}}$, (e) $\mathrm{Na\mathbf{B}H_4}$, (f) $\mathrm{H_2\mathbf{S_2}O_7}$, and (g) $\mathrm{KAl(\mathbf{S}O_4)_2\cdot 12H_2O}$.
(II) Write the chemical formulas for: Mercury(II) chloride, Nickel(II) sulphate, Tin(IV) oxide, Thallium(I) sulphate, Iron(III) sulphate, and Chromium(III) oxide.

Solution:

- **(I) Oxidation Numbers:**
  - (a) $\mathrm{NaH_2PO_4}$: $(+1)+2(+1)+x+4(-2)=0 \implies x=\mathbf{+5}$.
  - (b) $\mathrm{H_4P_2O_7}$: $4(+1)+2x+7(-2)=0 \implies 2x=+10 \implies x=\mathbf{+5}$.
  - (c) $\mathrm{K_2MnO_4}$: $2(+1)+x+4(-2)=0 \implies x=\mathbf{+6}$.
  - (d) $\mathrm{CaO_2}$ (calcium peroxide): $(+2)+2x=0 \implies x=\mathbf{-1}$.
  - (e) $\mathrm{NaBH_4}$ (metal hydride, $\mathrm{H}=-1$): $(+1)+x+4(-1)=0 \implies x=\mathbf{+3}$.
  - (f) $\mathrm{H_2S_2O_7}$ (oleum, no peroxide bond): $2(+1)+2x+7(-2)=0 \implies x=\mathbf{+6}$.
  - (g) $\mathrm{KAl(SO_4)_2\cdot 12H_2O}$: Water of crystallisation is neutral (0); in each $\mathrm{SO_4^{2-}}$ ion, $x+4(-2)=-2 \implies x=\mathbf{+6}$.
- **(II) Formulas from Stock Notation:** Mercury(II) chloride = $\mathrm{HgCl_2}$; Nickel(II) sulphate = $\mathrm{NiSO_4}$; Tin(IV) oxide = $\mathrm{SnO_2}$; Thallium(I) sulphate = $\mathrm{Tl_2SO_4}$; Iron(III) sulphate = $\mathrm{Fe_2(SO_4)_3}$; Chromium(III) oxide = $\mathrm{Cr_2O_3}$.

Worked Example 2: Contrasting Reactivity of $\mathrm{Pb_3O_4}$ with HCl vs $\mathrm{HNO_3}$

Problem: Why do the following two reactions of red lead ($\mathrm{Pb_3O_4}$) proceed differently?

$$\mathrm{Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + Cl_2 + 4H_2O}$$
$$\mathrm{Pb_3O_4 + 4HNO_3 \rightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O}$$

Solution: $\mathrm{Pb_3O_4}$ is a stoichiometric mixed oxide consisting of $2\,\mathrm{mol}$ of basic PbO (Pb in $+2$ state) and $1\,\mathrm{mol}$ of $\mathrm{PbO_2}$ (Pb in $+4$ state, a strong oxidising agent).
1. **With HCl:** $\mathrm{2PbO}$ undergoes a simple acid-base reaction ($\mathrm{2PbO + 4HCl \rightarrow 2PbCl_2 + 2H_2O}$), while $\mathrm{PbO_2}$ oxidises $\mathrm{Cl^-}$ (-1) of HCl to $\mathrm{Cl_2}$ (0) in a redox reaction ($\mathrm{PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + 2H_2O}$).
2. **With $\mathrm{HNO_3}$:** Because $\mathrm{HNO_3}$ is itself an oxidising agent (with N in its highest $+5$ state), no redox reaction occurs between $\mathrm{PbO_2}$ and $\mathrm{HNO_3}$. Only the acid-base reaction between $\mathrm{2PbO}$ and $\mathrm{4HNO_3}$ takes place, leaving unreacted $\mathrm{PbO_2}$ behind.

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Worked Example 3: Dual vs Pure Oxidising Behaviour, Isotopic Tracer Writing, & Special Reagents

Problem: Explain the following chemical observations:
(a) Why can $\mathrm{SO_2}$ and $\mathrm{H_2O_2}$ act as both oxidising and reducing agents, whereas $\mathrm{O_3}$ and $\mathrm{HNO_3}$ act only as oxidants?
(b) Why does thiosulphate ($\mathrm{S_2O_3^{2-}}$) react with $\mathrm{I_2}$ to give tetrathionate ($\mathrm{S_4O_6^{2-}}$) but with $\mathrm{Br_2}$ to give sulphate ($\mathrm{SO_4^{2-}}$)?
(c) What can we conclude from $\mathrm{XeO_6^{4-}(aq) + 2F^-(aq) + 6H^+(aq) \rightarrow XeO_3(g) + F_2(g) + 3H_2O(l)}$ and from the instability of $\mathrm{AgF_2}$?
(d) Why is photosynthesis written with $\mathrm{12H_2O}$ on the left and $\mathrm{6H_2O}$ on the right, and $\mathrm{O_3 + H_2O_2 \rightarrow H_2O + O_2 + O_2}$ written with two separate $\mathrm{O_2}$ molecules?

Solution:

- **(a)** In $\mathrm{SO_2}$, S is in the intermediate $+4$ state (can decrease to -2 or 0, or increase to $+6$). In $\mathrm{H_2O_2}$, O is in the intermediate -1 state (can decrease to -2 in $\mathrm{H_2O}$ or increase to 0 in $\mathrm{O_2}$). In $\mathrm{HNO_3}$, N is in its highest $+5$ state and can only decrease. Similarly, $\mathrm{O_3}$ only undergoes reduction in its reactions.
- **(b)** Bromine ($\mathrm{Br_2}$, $E^\circ = +1.09\,\mathrm{V}$) is a much stronger oxidising agent than iodine ($\mathrm{I_2}$, $E^\circ = +0.54\,\mathrm{V}$). Thus, $\mathrm{Br_2}$ oxidises S in $\mathrm{S_2O_3^{2-}}$ from $+2$ all the way to $+6$ in $\mathrm{SO_4^{2-}}$, whereas weaker $\mathrm{I_2}$ oxidises it only to an average state of $+2.5$ in $\mathrm{S_4O_6^{2-}}$.
- **(c)** In $\mathrm{XeO_6^{4-}}$ (from $\mathrm{Na_4XeO_6}$), Xe is in $+8$ state and oxidises $\mathrm{F^-}$ (-1) to $\mathrm{F_2}$ (0). This proves that $\mathrm{Na_4XeO_6}$ (perxenate) is an even stronger oxidising agent than $\mathrm{F_2}$! In $\mathrm{AgF_2}$, silver is in an unstable $+2$ state and readily accepts an electron to attain its stable $+1$ state ($\mathrm{Ag^{2+} + e^- \rightarrow Ag^+}$), acting as a very strong oxidant.
- **(d)** Using the **isotopic tracer technique** (with $^{18}\mathrm{O}$), it is shown that in photosynthesis ($\mathrm{6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2}$), all $\mathrm{6O_2}$ come exclusively from the $\mathrm{12H_2O}$ molecules. In $\mathrm{O_3 + H_2O_2 \rightarrow H_2O + O_2 + O_2}$, one $\mathrm{O_2}$ molecule comes from the reduction of $\mathrm{O_3}$ and the other $\mathrm{O_2}$ comes from the oxidation of $\mathrm{H_2O_2}$.

Worked Example 4: Stoichiometry in Ostwald's Process & Electrolysis Products

Problem: (I) In the first step of Ostwald's process, $\mathrm{4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)}$, what is the maximum weight of nitric oxide (NO) obtained starting with $10.00\,\mathrm{g}$ of $\mathrm{NH_3}$ and $20.00\,\mathrm{g}$ of $\mathrm{O_2}$?
(II) Predict the products of electrolysis of: (i) $\mathrm{AgNO_3(aq)}$ with Ag electrodes, (ii) $\mathrm{AgNO_3(aq)}$ with Pt electrodes, (iii) dilute $\mathrm{H_2SO_4(aq)}$ with Pt electrodes, and (iv) $\mathrm{CuCl_2(aq)}$ with Pt electrodes.

Solution:

- **(I) Limiting Reagent Calculation:** Molar masses: $\mathrm{NH_3} = 17\,\mathrm{g\,mol^{-1}}$, $\mathrm{O_2} = 32\,\mathrm{g\,mol^{-1}}$, $\mathrm{NO} = 30\,\mathrm{g\,mol^{-1}}$.
  Moles of $\mathrm{NH_3} = \frac{10.00}{17} = 0.588\,\mathrm{mol}$; Moles of $\mathrm{O_2} = \frac{20.00}{32} = 0.625\,\mathrm{mol}$.
  According to stoichiometry, $4\,\mathrm{mol\ NH_3}$ require $5\,\mathrm{mol\ O_2}$. Thus, $0.625\,\mathrm{mol\ O_2}$ reacts with $\frac{4}{5} \times 0.625 = 0.500\,\mathrm{mol\ NH_3}$. Hence, $\mathrm{O_2}$ **is the limiting reagent**.
  Moles of NO formed $= \frac{4}{5} \times 0.625 = 0.500\,\mathrm{mol}$. Mass of $\mathrm{NO} = 0.500\,\mathrm{mol} \times 30\,\mathrm{g\,mol^{-1}} = \mathbf{15.00}\,\mathrm{g}$.
- **(II) Electrolysis Products:**
  - (i) $\mathrm{AgNO_3(aq)}$ with Ag electrodes: Cathode: $\mathrm{Ag(s)}$ deposited; Anode: Ag anode dissolves to form $\mathrm{Ag^+(aq)}$.
  - (ii) $\mathrm{AgNO_3(aq)}$ with Pt electrodes: Cathode: $\mathrm{Ag(s)}$ deposited; Anode: $\mathrm{H_2O}$ oxidised to give $\mathrm{O_2(g)}$ (since $\mathrm{NO_3^-}$ cannot be oxidised further).
  - (iii) Dilute $\mathrm{H_2SO_4(aq)}$ with Pt electrodes: Cathode: $\mathrm{H_2(g)}$; Anode: $\mathrm{O_2(g)}$.
  - (iv) $\mathrm{CuCl_2(aq)}$ with Pt electrodes: Cathode: $\mathrm{Cu(s)}$ deposited; Anode: $\mathrm{Cl_2(g)}$ evolved.

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9. NEET UG Practice Questions with Complete Worked Solutions

Note: The following original practice questions are strictly aligned with the NEET UG examination pattern and difficulty level, including Single-Correct MCQs, Multi-Statement evaluations, Assertion-Reason, and Matrix-Match formats.

Practice Question 1 [Single-Correct MCQ — Oxidation Numbers & Structure]

What are the individual oxidation numbers of the three bromine atoms in tribromooctaoxide (Br_3O_8) and the four sulphur atoms in the tetrathionate ion ($S_4O_6^{2-}$), respectively?

(1) Br_3O_8 : $+6, +4, +6$; $S_4O_6^{2-}$: $+5, 0, 0, +5$

(2) Br_3O_8 : $+5, +6, +5$; $S_4O_6^{2-}$: $+6, 0, 0, +4$

(3) Br_3O_8 : $+6, +6, +4$; $S_4O_6^{2-}$: $+4, +1, +1, +4$

(4) Br_3O_8 : $+16/3$ for each Br; $S_4O_6^{2-}$: $+2.5$ for each S

Correct Answer: (1)

Worked Solution: In Br_3O_8, the two terminal bromine atoms are each bonded to three oxygen atoms via double bonds ($O=Br(=O)=O$), giving each terminal Br an oxidation state of $+6$, while the central Br atom is double-bonded to two oxygen atoms and single-bonded to the two Br atoms, giving it an oxidation state of $+4$ (average $= +16/3$). In $S_4O_6^{2-}$ ($^-O-SO_2-S-S-SO_2-O^-$), the two central sulphur atoms are bonded only to other sulphur atoms and have an oxidation state of 0, whereas the two terminal sulphur atoms each have an oxidation state of $+5$ (average $= +2.5$).

Practice Question 2 [Single-Correct MCQ — Disproportionation Reactions]

Which of the following oxoanions of chlorine **cannot** undergo a disproportionation reaction under any condition?

(1) ClO^-

(2) ClO_2^-

(3) ClO_3^-

(4) ClO_4^-

Correct Answer: (4)

Worked Solution: For a species to disproportionate, the central element must be in an intermediate oxidation state so that it can simultaneously increase and decrease its oxidation number. The oxidation states of Cl in ClO^-, ClO_2^-, ClO_3^-, and ClO_4^- are $+1$, $+3$, $+5$, and $+7$, respectively. Because $+7$ is the highest possible oxidation state for chlorine (Group 17, $17-10=+7$), ClO_4^- cannot be oxidised further and therefore does not disproportionate.

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Practice Question 3 [Single-Correct MCQ — Balancing Redox Equations]

Consider the balanced ionic equation for the oxidation of sulphite ions (SO_3^{2-}) by dichromate ions ($Cr_2O_7^{2-}$) in an acidic medium:

$$a\,Cr_2O_7^{2-}(aq) + b\,SO_3^{2-}(aq) + c\,H^+(aq) \rightarrow d\,Cr^{3+}(aq) + e\,SO_4^{2-}(aq) + f\,H_2O(l)$$

The stoichiometric coefficients a, b, c, and f in the lowest whole-number ratio are, respectively:

(1) $1, 3, 14, 7$

(2) $1, 3, 8, 4$

(3) $2, 3, 8, 4$

(4) $1, 6, 14, 7$

Correct Answer: (2)

Worked Solution: In $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$, each Cr decreases from $+6$ to $+3$, giving a total decrease of $2 \times 3 = 6$ units per $Cr_2O_7^{2-}$ ion. In $SO_3^{2-} \rightarrow SO_4^{2-}$, S increases from $+4$ to $+6$, giving an increase of 2 units per SO_3^{2-} ion. Multiplying SO_3^{2-} and SO_4^{2-} by 3 equalises the oxidation number change:

$$Cr_2O_7^{2-} + 3SO_3^{2-} \rightarrow 2Cr^{3+} + 3SO_4^{2-}$$

Left side charge $= -2 + 3(-2) = -8$; right side charge $= 2(+3) + 3(-2) = 0$. Adding $8H^+$ to the left side balances the charge, and adding $4H_2O$ to the right side balances hydrogen and oxygen:

$$1\,Cr_2O_7^{2-}(aq) + 3\,SO_3^{2-}(aq) + 8\,H^+(aq) \rightarrow 2\,Cr^{3+}(aq) + 3\,SO_4^{2-}(aq) + 4\,H_2O(l)$$

Thus, $a = 1, b = 3, c = 8$, and $f = 4$.

Practice Question 4 [Single-Correct MCQ — Balancing in Basic Medium]

In a basic medium, permanganate ion (MnO_4^-) oxidises iodide ion (I^-) to produce molecular iodine (I_2) and manganese(IV) oxide (MnO_2). How many moles of OH^- ions are produced per mole of I_2 formed in the balanced equation?

(1) $8\,mol$

(2) $4\,mol$

(3) $\frac{8}{3}\,mol$

(4) $\frac{4}{3}\,mol$

Correct Answer: (3)

Worked Solution: Balancing the two half-reactions in basic medium:
Oxidation half: $2I^-(aq) \rightarrow I_2(s) + 2e^-$ (multiply by 3)
Reduction half: $MnO_4^-(aq) + 2H_2O(l) + 3e^- \rightarrow MnO_2(s) + 4OH^-(aq)$ (multiply by 2)
Overall balanced equation:

$$6I^-(aq) + 2MnO_4^-(aq) + 4H_2O(l) \rightarrow 3I_2(s) + 2MnO_2(s) + 8OH^-(aq)$$

For $3\,mol$ of I_2 formed, $8\,mol$ of OH^- are produced. Therefore, per $1\,mol$ of I_2 formed, $\frac{8}{3}\,mol$ of OH^- are produced.

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Practice Question 5 [Assertion-Reason — Non-Redox Decomposition]

Assertion (A): The thermal decomposition of potassium chlorate ($2\mathrm{KClO_3(s)} \xrightarrow{\Delta} 2\mathrm{KCl(s)} + 3\mathrm{O_2(g)}$) is a decomposition redox reaction, whereas the thermal decomposition of calcium carbonate ($\mathrm{CaCO_3(s)} \xrightarrow{\Delta} \mathrm{CaO(s)} + \mathrm{CO_2(g)}$) is not a redox reaction.
Reason (R): Every decomposition reaction yields at least one product in the free elemental state.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (3)

Worked Solution: Assertion (A) is true: in $2\mathrm{KClO_3} \rightarrow 2\mathrm{KCl} + 3\mathrm{O_2}$, Cl is reduced from $+5$ to -1 and O is oxidised from -2 to 0, whereas in $\mathrm{CaCO_3} \rightarrow \mathrm{CaO} + \mathrm{CO_2}$, the oxidation numbers of Ca ($+2$), C ($+4$), and O (-2) remain unchanged. Reason (R) is false because not all decomposition reactions produce a free element (as demonstrated by $\mathrm{CaCO_3} \rightarrow \mathrm{CaO} + \mathrm{CO_2}$); only *redox* decomposition reactions require at least one product in the elemental state.

Practice Question 6 [Multi-Statement Evaluation — Halogens & Alkaline Medium]

Examine the following statements regarding the redox behaviour of halogens:

I. Among halogens, oxidising power decreases down the group from $\mathrm{F_2}$ to $\mathrm{I_2}$, while among hydrohalic acids, reducing power increases from HF to HI.

II. Fluorine reacts with cold dilute alkali to undergo disproportionation into $\mathrm{F^-}$ and $\mathrm{FO^-}$.

III. Chlorine reacts with aqueous $\mathrm{OH^-}$ to form $\mathrm{Cl^-}$ and $\mathrm{ClO^-}$, where $\mathrm{ClO^-}$ acts as the active bleaching species in household bleach.

IV. $\mathrm{F^-}$ ions cannot be oxidised to $\mathrm{F_2}$ by any chemical oxidising agent in aqueous solution and must be oxidised electrolytically.

Which of the above statements are **correct**?

(1) I, II, and III only

(2) I, III, and IV only

(3) II and IV only

(4) I, II, III, and IV

Correct Answer: (2)

Worked Solution: Statement I is correct (HI readily loses electrons to form $\mathrm{I_2}$, even reducing $\mathrm{H_2SO_4}$, whereas $\mathrm{F_2}$ has $E^\circ = +2.87\ \mathrm{V}$). Statement II is false: because fluorine is the most electronegative element, it never exhibits a positive oxidation state and *never* disproportionates; with $\mathrm{OH^-}$ it forms $\mathrm{F^-}$ (-1) and $\mathrm{OF_2}$ ($\mathrm{F} = -1, \mathrm{O} = +2$). Statements III and IV are both correct.

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Practice Question 7 [Single-Correct MCQ — Standard Electrode Potentials & Reducing Power]

Given the standard electrode potentials at $298\,\mathrm{K}$:
$\mathrm{K^+/K} = -2.93\,\mathrm{V}$, $\mathrm{Ag^+/Ag} = +0.80\,\mathrm{V}$, $\mathrm{Hg^{2+}/Hg} = +0.79\,\mathrm{V}$, $\mathrm{Mg^{2+}/Mg} = -2.37\,\mathrm{V}$, $\mathrm{Cr^{3+}/Cr} = -0.74\,\mathrm{V}$.
Arrange these metals in the **increasing order of their reducing power**:

(1) $\mathrm{K < Mg < Cr < Hg < Ag}$

(2) $\mathrm{Ag < Hg < Cr < Mg < K}$

(3) $\mathrm{Hg < Ag < Cr < Mg < K}$

(4) $\mathrm{Ag < Cr < Hg < Mg < K}$

Correct Answer: (2)

Worked Solution: A more negative (or less positive) standard reduction potential (E°) indicates a greater tendency of the metal to lose electrons and act as a stronger reducing agent. Since E° decreases in the order $\mathrm{Ag}\ (+0.80\,\mathrm{V}) > \mathrm{Hg}\ (+0.79\,\mathrm{V}) > \mathrm{Cr}\ (-0.74\,\mathrm{V}) > \mathrm{Mg}\ (-2.37\,\mathrm{V}) > \mathrm{K}\ (-2.93\,\mathrm{V})$, the reducing power of the metals increases in the exact order: $\mathrm{Ag < Hg < Cr < Mg < K}$.

Practice Question 8 [Single-Correct MCQ — Feasibility of Redox Reactions using E°]

Using the standard reduction potentials: $E^\circ_{\mathrm{Br_2/Br^-}} = +1.09\,\mathrm{V}$, $E^\circ_{\mathrm{Ag^+/Ag}} = +0.80\,\mathrm{V}$, $E^\circ_{\mathrm{Fe^{3+}/Fe^{2+}}} = +0.77\,\mathrm{V}$, $E^\circ_{\mathrm{I_2/I^-}} = +0.54\,\mathrm{V}$, and $E^\circ_{\mathrm{Cu^{2+}/Cu}} = +0.34\,\mathrm{V}$, identify which of the following reactions is **NOT spontaneous (not feasible)** under standard conditions:

(1) $\mathrm{2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)}$

(2) $\mathrm{2Ag^+(aq) + Cu(s) \rightarrow 2Ag(s) + Cu^{2+}(aq)}$

(3) $\mathrm{Fe^{3+}(aq) + Ag(s) \rightarrow Fe^{2+}(aq) + Ag^+(aq)}$

(4) $\mathrm{Br_2(aq) + 2Fe^{2+}(aq) \rightarrow 2Br^-(aq) + 2Fe^{3+}(aq)}$

Correct Answer: (3)

Worked Solution: A redox reaction is feasible if the species being reduced has a higher E° than the species being oxidised ($E^\circ_{\mathrm{cell}} = E^\circ_{\mathrm{cathode}} - E^\circ_{\mathrm{anode}} > 0$):

- (1) $E^\circ_{\mathrm{cell}} = 0.77 - 0.54 = +0.23\,\mathrm{V} > 0$ (Feasible).
- (2) $E^\circ_{\mathrm{cell}} = 0.80 - 0.34 = +0.46\,\mathrm{V} > 0$ (Feasible).
- (3) $E^\circ_{\mathrm{cell}} = E^\circ_{\mathrm{Fe^{3+}/Fe^{2+}}} - E^\circ_{\mathrm{Ag^+/Ag}} = 0.77 - 0.80 = -0.03\,\mathrm{V} < 0$ (**Not feasible**).
- (4) $E^\circ_{\mathrm{cell}} = 1.09 - 0.77 = +0.32\,\mathrm{V} > 0$ (Feasible).

BOSONANCE

Practice Question 9 [Matrix-Match — Redox Titrations & Indicators]

Match the redox system or reagent in **Column I** with its characteristic role or indicator behaviour in **Column II**:

Column I (Reagent / Titration System)	Column II (Indicator / Characteristic Role)
(A) $\mathrm{MnO_4^-}$ against $\mathrm{Fe^{2+}}$ or $\mathrm{C_2O_4^{2-}}$	(p) Oxidises $\mathrm{S_2O_3^{2-}}$ completely to $\mathrm{SO_4^{2-}}$
(B) $\mathrm{Cr_2O_7^{2-}}$ titration of $\mathrm{Fe^{2+}}$	(q) Acts as a self-indicator with pink end point at $10^{-6}\ \mathrm{mol\,L^{-1}}$
(C) Estimation of $\mathrm{Cu^{2+}}$ using KI and $\mathrm{Na_2S_2O_3}$	(r) Oxidises diphenylamine indicator just after the equivalence point
(D) Reaction of $\mathrm{Br_2(l)}$ with $\mathrm{S_2O_3^{2-}(aq)}$	(s) Starch gives an intense blue complex with liberated $\mathrm{I_2}$

(1) A–q, B–r, C–s, D–p

(2) A–r, B–q, C–s, D–p

(3) A–q, B–s, C–r, D–p

(4) A–q, B–r, C–p, D–s

Correct Answer: (1)

Worked Solution: (A) $\mathrm{MnO_4^-}$ is intensely coloured and acts as a self-indicator (q). (B) $\mathrm{Cr_2O_7^{2-}}$ is not a self-indicator; it oxidises diphenylamine to produce an intense blue colour at the end point (r). (C) $\mathrm{Cu^{2+}}$ oxidises $\mathrm{I^-}$ to $\mathrm{I_2}$, which forms a deep blue complex with starch and is titrated with $\mathrm{S_2O_3^{2-}}$ to $\mathrm{S_4O_6^{2-}}$ (s). (D) $\mathrm{Br_2}$, being a stronger oxidant than $\mathrm{I_2}$, oxidises $\mathrm{S_2O_3^{2-}}$ to $\mathrm{SO_4^{2-}}$ (p).

Practice Question 10 [Single-Correct MCQ — Selective Oxidising Agents in Organic & Inorganic Systems]

Consider the reactions of hypophosphorous acid ($\mathrm{H_3PO_2}$) and benzaldehyde ($\mathrm{C_6H_5CHO}$) with silver(I) and copper(II) reagents:
I. Both $\mathrm{AgNO_3(aq)}$ and $\mathrm{CuSO_4(aq)}$ oxidise $\mathrm{H_3PO_2}$ to $\mathrm{H_3PO_4}$ while getting reduced to $\mathrm{Ag(s)}$ and $\mathrm{Cu(s)}$.
II. Ammoniacal silver(I) ion, $\mathrm{[Ag(NH_3)_2]^+}$, in basic medium oxidises $\mathrm{C_6H_5CHO}$ to $\mathrm{C_6H_5COO^-}$, whereas alkaline $\mathrm{Cu^{2+}}$ solution shows no change with $\mathrm{C_6H_5CHO}$.
What inference can be drawn from these observations?

(1) $\mathrm{Cu^{2+}}$ is a stronger oxidising agent than $\mathrm{Ag^+}$ in both acidic and basic media.

(2) Both $\mathrm{Ag^+}$ and $\mathrm{Cu^{2+}}$ act as oxidising agents, but $\mathrm{Ag^+}$ is a stronger oxidising agent than $\mathrm{Cu^{2+}}$.

(3) $\mathrm{H_3PO_2}$ acts as an oxidising agent towards $\mathrm{AgNO_3}$ and $\mathrm{CuSO_4}$.

(4) $\mathrm{C_6H_5CHO}$ reduces $\mathrm{Cu^{2+}}$ to $\mathrm{Cu_2O}$ rapidly at room temperature.

Correct Answer: (2)

Worked Solution: Both $\mathrm{Ag^+}$ and $\mathrm{Cu^{2+}}$ oxidise the strong reducing agent $\mathrm{H_3PO_2}$ (P in $+1$ state) to $\mathrm{H_3PO_4}$ (P in $+5$ state). However, with benzaldehyde ($\mathrm{C_6H_5CHO}$), only $\mathrm{[Ag(NH_3)_2]^+}$ is strong enough to oxidise it to benzoate ion ($\mathrm{C_6H_5COO^-}$), whereas $\mathrm{Cu^{2+}}$ in basic medium cannot oxidise $\mathrm{C_6H_5CHO}$. This confirms that while both act as oxidants, $\mathrm{Ag^+}$ ($E^\circ = +0.80\ \mathrm{V}$) is a stronger oxidising agent than $\mathrm{Cu^{2+}}$ ($E^\circ = +0.34\ \mathrm{V}$).

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Practice Question 11 [Single-Correct MCQ — Pseudohalogens & Disproportionation]

Cyanogen gas, $(CN)_2(g)$, reacts with aqueous hydroxide ions according to the equation:

$$(CN)_2(g) + 2OH^-(aq) \rightarrow CN^-(aq) + CNO^-(aq) + H_2O(l)$$

Which of the following statements is **incorrect** regarding this reaction?

(1) Cyanogen behaves as a pseudohalogen, mimicking the reaction of $Cl_2(g)$ with cold dilute $OH^-(aq)$.

(2) The reaction is a disproportionation redox reaction.

(3) Taking the oxidation state of nitrogen as -3, the oxidation state of carbon changes from $+3$ in $(CN)_2$ to $+2$ in CN^- and $+4$ in CNO^-.

(4) Cyanogen acts exclusively as an oxidising agent in this reaction.

Correct Answer: (4)

Worked Solution: Since nitrogen is more electronegative than carbon and retains an oxidation state of -3 throughout, carbon has an oxidation state of $+3$ in $(CN)_2$, $+2$ in CN^- (reduction), and $+4$ in CNO^- (oxidation). Therefore, $(CN)_2$ undergoes disproportionation and acts as *both* an oxidising agent and a reducing agent, making statement (4) incorrect.

Practice Question 12 [Single-Correct MCQ — Galvanic Cell & Electrode Polarity]

In a galvanic cell where the net reaction $Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$ takes place, which of the following statements is **correct**?

(1) The silver electrode is negatively charged and electrons flow from Ag to Zn in the external circuit.

(2) The zinc electrode is the negatively charged anode, electrons flow from Zn to Ag in the external wire, and conventional current flows from Ag to Zn.

(3) Electrons flow through the salt bridge from the $AgNO_3$ solution to the $ZnSO_4$ solution.

(4) Oxidation occurs at the silver electrode: $Ag^+(aq) + e^- \rightarrow Ag(s)$.

Correct Answer: (2)

Worked Solution: Zinc is oxidised ($Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$), releasing electrons onto the zinc rod and making the Zn anode negatively charged. These electrons travel through the external circuit from the Zn electrode to the Ag cathode, where $Ag^+(aq)$ ions are reduced ($Ag^+(aq) + e^- \rightarrow Ag(s)$). Conventional current flows opposite to electron flow (from Ag to Zn), and ions (not electrons) carry current inside the cell across the salt bridge.

10. Common Pitfalls & Quick Revision Checklist

Common Exam Pitfalls to Avoid

- **Blindly applying the algebraic formula to peroxy compounds:** Never report oxidation numbers higher than the group valence (e.g., $+8$ for S in H_2SO_5 or $+10$ for Cr in CrO_5). Always account for peroxide ($-O-O-$) oxygens at -1 each; both S in H_2SO_5 and Cr in CrO_5 are in the $+6$ state.
- **Assuming all decomposition reactions are redox:** Check whether at least one elemental substance is formed. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ has no change in oxidation state and is *not* a redox reaction.

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- **Assuming fluorine disproportionates in alkali:** Unlike $\mathrm{Cl_2}$, $\mathrm{Br_2}$, and $\mathrm{I_2}$, fluorine ($\mathrm{F_2}$) *never* exhibits positive oxidation states and cannot disproportionate. With $\mathrm{OH^-}$ it forms $\mathrm{F^-}$ and $\mathrm{OF_2}$; with ice ($\mathrm{H_2O(s) + F_2(g) \rightarrow HF(g) + HOF(g)}$), F is -1 in both HF and HOF while O is oxidised from -2 to 0 in HOF.
- **Confusing average oxidation number with actual atomic oxidation states:** In $\mathrm{C_3O_2}$ $(+2, 0, +2)$, $\mathrm{Br_3O_8}$ ($+6, +4, +6$), and $\mathrm{S_4O_6^{2-}}$ $(+5, 0, 0, +5)$, fractional values are only averages.
- **Reversing the sign interpretation of E°:** A more *negative* E° means a stronger *reducing* agent (Li is the strongest reducing agent in aqueous solution with $E^\circ = -3.05\ \mathrm{V}$), whereas a more *positive* E° means a stronger *oxidising* agent ($\mathrm{F_2}$ has $E^\circ = +2.87\ \mathrm{V}$).

Concise Final Recap

- **Oxidation:** Addition of O/electronegative element, removal of H/electropositive element, **loss of electrons**, or **increase in oxidation number**.
- **Reduction:** Removal of O/electronegative element, addition of H/electropositive element, **gain of electrons**, or **decrease in oxidation number**.
- **Disproportionation:** Requires an element with at least three oxidation states starting in an intermediate state (e.g., $\mathrm{H_2O_2}, \mathrm{P_4}, \mathrm{S_8}, \mathrm{Cl_2}, \mathrm{Mn^{3+}}$ where $\mathrm{2Mn^{3+} + 2H_2O \rightarrow Mn^{2+} + MnO_2 + 4H^+}$).
- **Metal Displacement Order:**
  $\mathrm{Mg}\ (-2.36\ \mathrm{V}) > \mathrm{Al}\ (-1.66\ \mathrm{V}) > \mathrm{Zn}\ (-0.76\ \mathrm{V}) > \mathrm{Fe}\ (-0.44\ \mathrm{V}) > \mathrm{Cu}\ (+0.34\ \mathrm{V}) > \mathrm{Ag}\ (+0.80\ \mathrm{V})$
  .

BOSONANCE